UNIT 4

Pranshu

ROBOTICS

- The subject of robotics covers many different areas. Robots alone are hardly ever useful. They are used together with other devices, peripherals, and manufacturing machines.
- They are generally integrated into a system, which as a whole is designed to perform a task or to do an operation.
- They are capable of performing many different tasks and operations precisely and do not require common safety and comfort elements that humans needs.
- However, it takes much effort and many resources to make a robot function properly. As with humans, robots can do certain things, but not other things. As long as they are designed properly for the intended purpose, they are very useful and will continue to be used.
- Robotics is the art, knowledge base, and the know-how of designing, applying, and using robots in human endeavors.

HISTORY OF ROBOTS

- The word robot always refers to an automated multifunctional manipulator that works by energy, to perform a variety of tasks.
- The word robot was introduced in 1920 in a play by Karel Capek called R.U.R., or Rossum's Universal Robots.
- Robot comes from the Czech word robota, meaning forced labor or drudgery.
- One of the first types of robots was a feedback (self-correcting) control mechanism. It was a watering trough that used a float to sense the water level. When the water gets to low, the float drops, opens a valve, and more water dumps into the trough.
- As the water rises, so does the float. Once it reaches a certain height, the valve is closed and the water is shut off. In 1954, the American inventor George Devol, Jr. developed a primitive arm that could be programmed to perform specific tasks.

HISTORY OF ROBOTS

- In 1975, the American mechanical engineer Victor Scheinman developed a truly flexible multipurpose manipulator known as the Programmable Universal Manipulation Arm (PUMA).
- PUMA was capable of moving an object and placing it with any orientation in a desired location. Before the 1960s, robot usually meant a manlike mechanical device (mechanical man or humanoid) capable of performing human tasks or behaving in a human manner.
- Today robots come in all shapes and sizes, including small robots and larger wheeled robots that play football with a full-size ball. In 1995, there were about 700,000 robots operating in the world. A major user of robots is the automobile industry. General Motors uses approximately 16,000 robots for tasks such as spot welding, painting, machine loading, parts transfer, and assembly

DEFINITION OF A ROBOT

A robot is a computer-controlled machine that is programmed to move, manipulate objects, and accomplish work while interacting with its environment. According to the Robot Institute of America (1979), a robot is defined as "a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks."

Or

Industrial robot is defined as "a number of rigid links connected by joints of different types that are controlled and monitored by computer."

INDUSTRIAL ROBOT

Industrial robots are advanced automation systems, mainly controlled by a computer. Today computers form an important part of industrial automation.

They supervise production lines and control manufacturing systems (e.g., machine tools, welders, laser cutting devices, etc.).

The new generation of robots executes various tasks in industrial systems and they participate in the full automation of factories.

Japanese defined industrial robots in four levels:

- Manual manipulators: perform fixed or preset task sequences.
- Playbacks: repeat pre-programmed fixed instructions.
- NC robot: carry out tasks through numerically loaded information.
- Intelligent robots: perform through their own recognition capabilities.

INDUSTRIAL ROBOT

- A robot and a crane are very similar in the way they operate and in the way they are designed. Both possess a number of links attached serially to each other with joints, where some type of actuator can move each joint. In both systems, the hand of the manipulator can be moved in space and be placed in any desired location within the workspace of the system, each one can carry a certain amount of load and each one is controlled by a central controller which controls the actuators. However, one is called a robot and the other manipulator (or, in this case, a crane).
- The fundamental difference between the two is that in the case of a crane, a human operates and controls the actuators, whereas a computer that runs a program controls the robot manipulator.

LAWS OF ROBOTICS

Isaac Asimov proposed three laws of robotics and he later added a "zeroth law."

- Law One: A robot may not injure a human being, or, through inaction, allow a human being to come to harm, unless this would violate a higher order law.
- Law Two: A robot must obey orders given to it by human beings, except where such orders would conflict with a higher order law.
- Law Three: A robot must protect its own existence as long as such protection does not conflict with a higher order law.
- Law Zero: A robot may not injure humanity, or, through inaction, allow humanity to come to harm.

MOTIVATING FACTORS

For robotics systems to be introduced to the industrial world, they must have positive factors that would make a difference in using them.
The motivating factors can be categorized as:

- Technical factors
- Economic factors
- Social factors

1.Technical Factors

Robots can do incomparable tasks that humans can’t do. It is generally considered that humans can’t match the speed, quality, reliability, and the endurance of a robotic system. In that they offer:

- High flexibility of product type and variation.
- Lower preparation time than hard automation.
- Better quality of products.
- Fewer rejects and less waste than labor intensive production

2. Economic Factors

- The needs to increase production rates to remain competitive.
- Pressure from the market place to improve quality.
- Increasing costs. Shortage of skilled labor.

3. Social Factors

- Some people think that the use of robotized systems increases the unemployment of workers and prevents many people from a main income. But the usage of robots causes a reduction of workload on workers and prevents dangerous working conditions as robots can be used in hazardous environments.

ADVANTAGES AND DISADVANTAGES OF ROBOTS

- Robots offer specific benefits to workers, industries, and countries. If introduced correctly, industrial robots can improve the quality of life by freeing workers from dirty, boring, dangerous, and heavy labor.
- It can be said, therefore, that robots give the possibility to humans to occupy with jobs, that they can execute better.
- It is true that robots can cause unemployment by replacing human workers but robots also create jobs such as robot technicians, salesmen, engineers, programmers, and supervisors.

Advantages of Robots

- Increase in productivity, safety, efficiency, quality, and consistency of products with the use of robots.
- Robots can work in hazardous environments without the need for life support, comfort, or concern about safety.
- Robots need no environmental comfort such as lighting, air conditioning, ventilation, and noise protection.
- Robots can work continuously without experiencing fatigue or boredom, do not have hangovers, and need no medical insurance or vacation.
- Robots have repeatable precision at all times, unless something happens to them or unless they wear out.
- Robots can be much more accurate than human

Disadvantages of Robots

- Robots replace human workers creating economic problems, such as lost salaries, and social problems such as dissatisfaction and resentment among workers.
- Robots lack capability to respond in emergencies, unless the situation is predicted and the response is included in the system.
- Safety measures are needed to ensure that they do not injure operators and machines working with them.
- Robots are costly due to initial cost of equipment, installation costs, need for training, and need for programming

CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- An industrial robot has a hand, wrist, arm, base, lifting power, repeatability, manual control, automatic control, memory, programs, safety interlock, speed of operation, computer interface, reliability, and easy maintenance.
- The hand of a robot is known as a gripper or end effector or end-of-arm tooling. It is the driven mechanical device(s) attached to the end of the manipulator.
- The wrist of the robot is used to aim the hand at any part of the work piece. The wrist may have three motions: pitch (up-and-down motion), yaw (side-to-side motion), and roll (rotating motion).

CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- The arm is used to move the hand within reach of a part or work piece. It can pivot at its elbow and at its shoulder joint.
- The waist, or base, of the robot supports the arm and is called the shoulder. The arm can rotate about the shoulder.
- Repeatability is the replication of motion within some specified precision or tolerance.
- An RCC or remote center compliance device helps pull the hand or tool into the required position by acting as a multi-axis float.
- A manual control device is used to teach the robot how to do a new task.

CHARACTERISTICS OF AN INDUSTRIAL ROBOT

- An automatic control system is used to carry out the instructions stored in the robot's memory.
- The robot's memory holds a library of programs to use in executing different tasks.
- Safety interlocks prevent the robot from inserting a hand into a machine and causing damage to the robot and machine.
- The robot's speed of operation in performing a task should be at least equal to that of the human worker it is replacing.
- The robot's computer interface enables the robot to use the computer's larger memory to hold more task programs and to synchronize its actions with a complete production line of robots and other machines.

COMPONENTS OF AN INDUSTRIAL ROBOT

Industrial robot systems consist of four major subsystems as shown in Figure 13.1. Mechanical unit Drive Control system Tooling

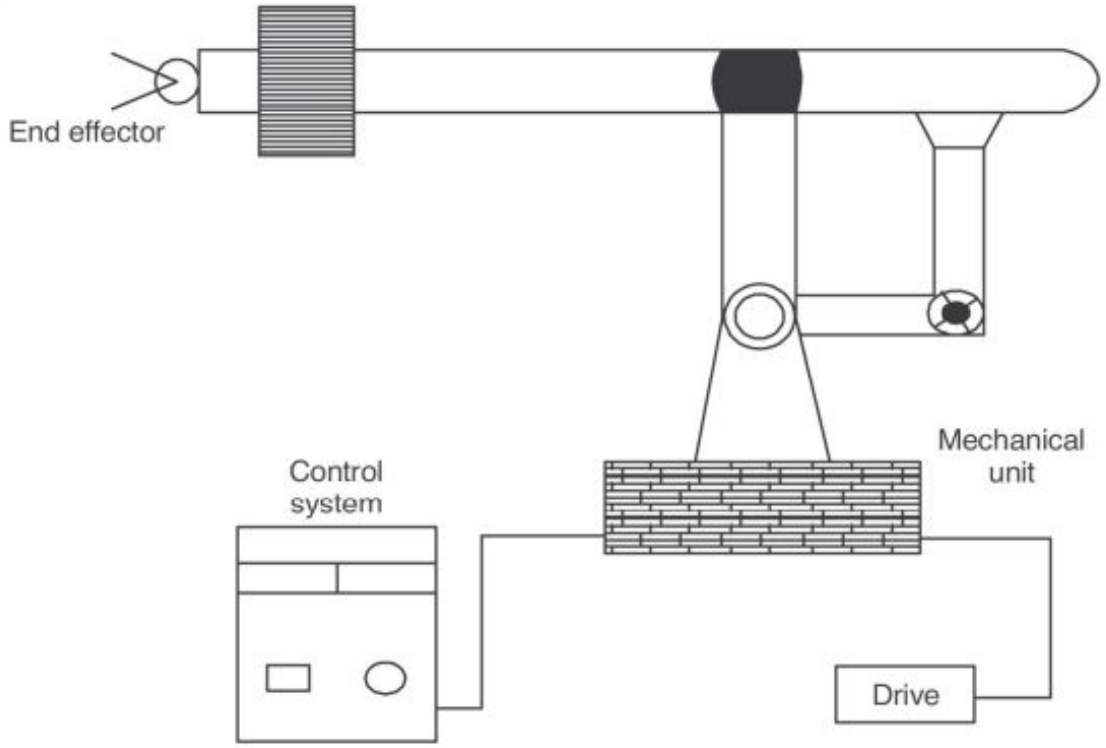


FIGURE 13.1 Components of an Industrial Robot.

COMPONENTS OF AN INDUSTRIAL ROBOT

- Mechanical Unit. The mechanical unit refers to the robot's manipulative arm and its base. Tooling such as end effectors, tool changers, and grippers are attached to the wrist-tooling interface. The mechanical unit consists of a fabricated structural frame with provisions for supporting mechanical linkage and joints, guides, actuators, control valves, limiting devices, and sensors. The physical dimensions, design, and loading capability of the robot depends upon the application requirements.

- Drive. An important component of the robot is the drive system. The drive system supplies the power, which enables the robot to move. Drive for a robot may be hydraulic, pneumatic, or electric. Hydraulic drives have been used for heavier lift systems. Pneumatic drives have been used for high speed, non-servo robots and are often used for powering tooling such as grippers. Electric drive systems can provide both lift and/or precision, depending on the motor and servo system selection and design. An AC or DC powered motor may be used depending on the system design and applications.

COMPONENTS OF AN INDUSTRIAL ROBOT

- Control System. Controller is the brain of the robot. Controller is a communication and information-processing device that initiates, terminates, and coordinates the motions and sequences of a robot. Most industrial robots incorporate computer or microprocessor based controllers. These perform computational functions and interface with sensors, grippers, tooling, and other peripheral equipment.
- Controller programming may be done on-line or from off-line control stations. Programs may be on cassettes, floppy disks, internal drives, or in memory; and may be loaded or downloaded by cassettes, disks, or telephone modem. Some robot controllers have sufficient computational ability, memory capacity, and input/output capability to serve as system controllers for other equipment and processes.

COMPONENTS OF AN INDUSTRIAL ROBOT

- Tooling. Tooling is manipulated by the robot to perform the functions required for the application. Depending on the application, the robot may have one functional capability, such as making spot welds or spray-painting.
- These capabilities may be integrated with the robot's mechanical system or may be attached at the robot's wrist-end effector interface. Alternatively, the robot may use multiple tools that may be changed manually (as part of set-up for a new program) or automatically during a work cycle.
- Tooling and objects that may be carried by a robot's gripper can significantly increase the envelope in which objects or humans may be struck.
- Tooling manipulated by the industrial robot and carried objects can cause more significant hazards than motion of the bare robotic system.
- The hazards added by the tooling should be addressed as part of the risk assessment.

COMPONENTS OF AN INDUSTRIAL ROBOT

- Sensors. Sensors are used to collect information about the internal state of the robot or to communicate with the outside environment. As in humans, the robot controller needs to know where each link of the robot is, in order to know the robot's configuration.
- The state of the human body is determined because feedback sensors in human's central nervous system embedded in their muscle tendons send information to the brain.
- The brain uses this information to determine the length of their muscles, and thus, the state of their arms, legs, etc. The same is true for robots; sensors integrated into the robot send information about each joint or link to the controller, which determines the configuration of the robot.
- Robots are often equipped with external sensory devices such as a vision system, touch and tactile sensors, speech synthesizers, etc., which enable the robot to communicate with the outside world.

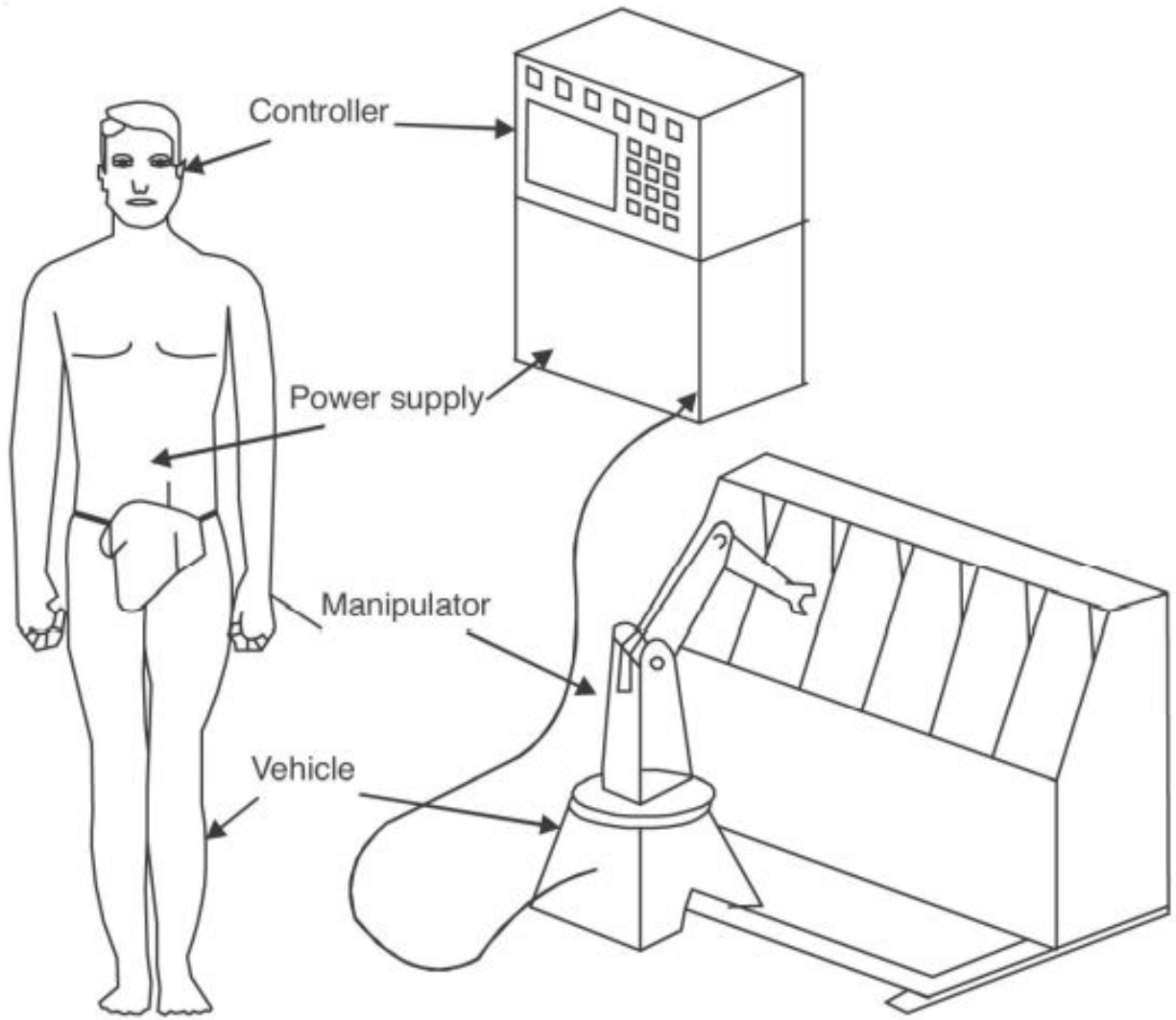
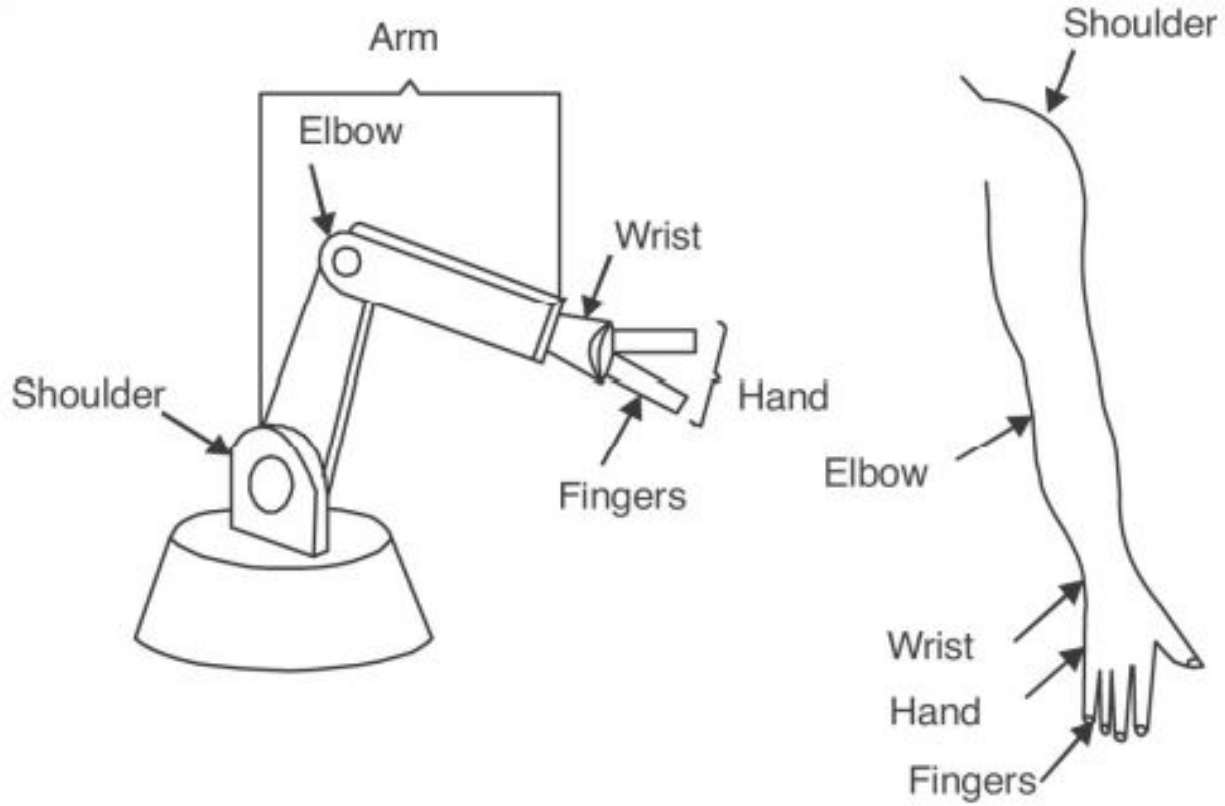


FIGURE 13.2 Comparison of Parts of Robot and Human Being.

COMPARISON OF THE HUMAN AND ROBOT MANIPULATOR

- The parts of a robot's manipulator are named after similar parts in a human's chief manipulators, i.e., the arm and the hand.



ROBOT WRIST AND END OF ARM TOOLS

- Six axis coordinates are required by a robot in order to completely specify the location and orientation of an object. Three coordinates can locate the center of gravity of an object (e.g., x, y, and z coordinates-in a rectangular coordinate system).
- Three other coordinates are normally achieved by adding wrist and hand movements with the end of arm tooling.
- There are three basic types of wrist motions:
  - Pitch - Rotational or bending movement in a vertical plane.
  - Yaw - Rotational or twisting movement in a horizontal plane.
  - Roll - Rotational or swivel movement.

ROBOT WRIST AND END OF ARM TOOLS

- Additionally, the wrist (refer to Figure 13.4) serves as the mounting point for a variety of devices. These end effectors are either hand, or gripping devices, or job specific tool

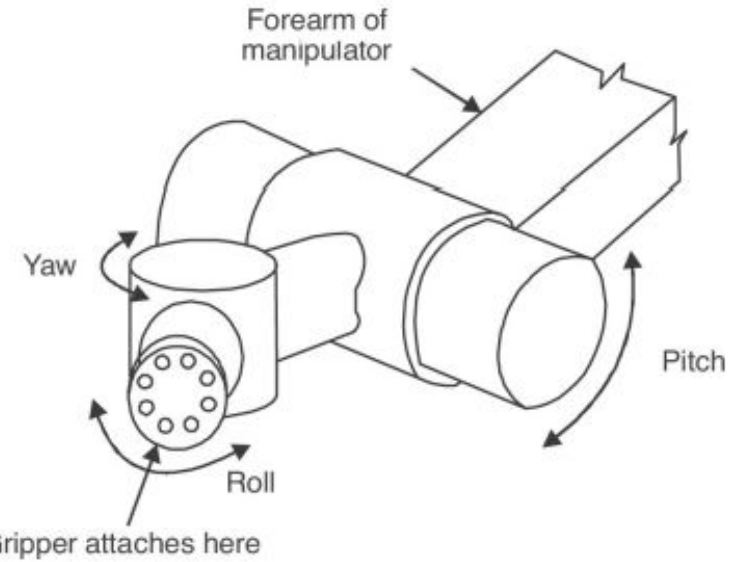


FIGURE 13.4 Robotic Wrist.

ROBOT WRIST AND END OF ARM TOOLS

- Figure 13.5 shows the various arms and wrist motions of a robot.

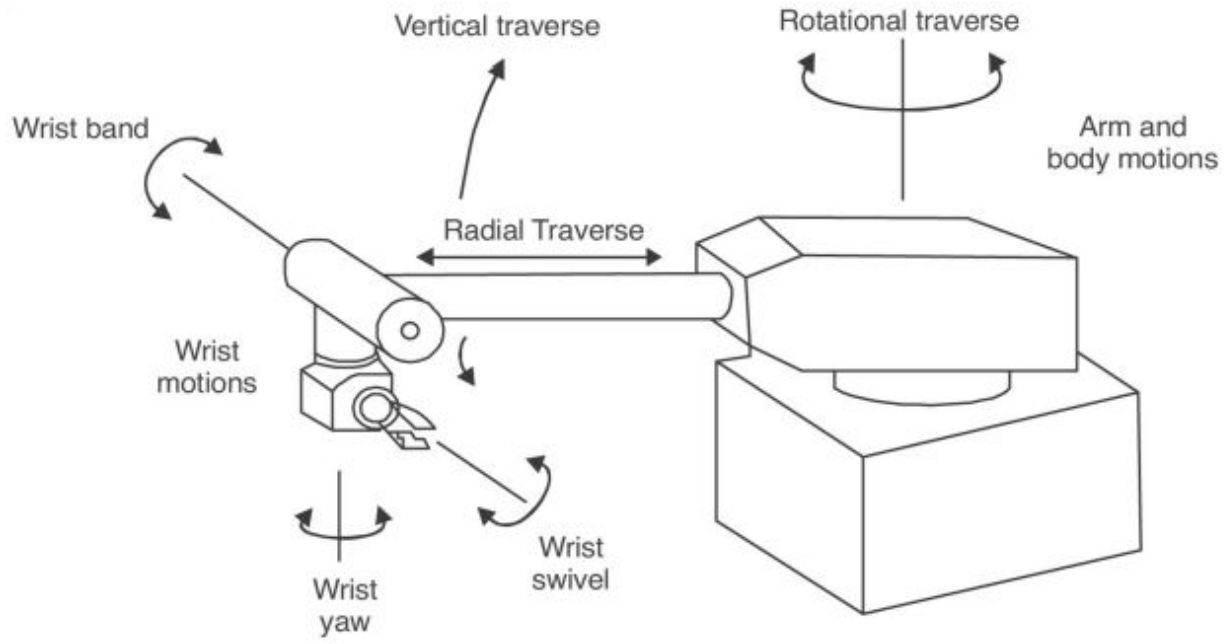


FIGURE 13.5 Arm and Wrist Motions of a Robot.

ROBOT WRIST AND END OF ARM TOOLS

- End Effector. The end effector is the device at the end of the robot arm. The end effector attached to the robot wrist is shown in Figure 13.6.

There are two main types of end effectors: grippers and tools.

i. Grippers: Grippers are devices, which can be used for holding or gripping an object. These include mechanical hands and anything like hooks, magnets, and suction devices, which can be used for holding or gripping.

ii. Tools: Tools are devices, which robots use to perform operations on an object, e.g., drills, paint sprays, grinders, welding torches, and any other tool which get a specific job done.

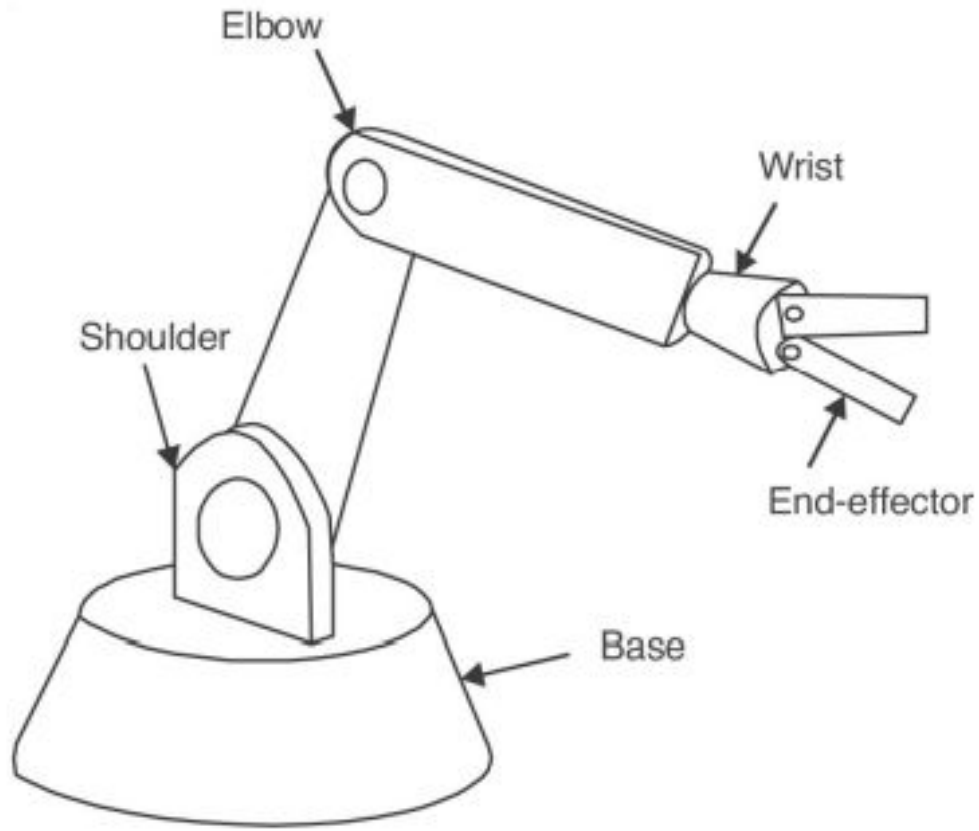


FIGURE 13.6 End Effector Attached to Robot Wrist.

ROBOT TERMINOLOGY

There is a set of basic terminology and concepts common to all robots. These terms follow with brief explanations of each.

ROBOT TERMINOLOGY There is a set of basic terminology and concepts common to all robots. These terms follow with brief explanations of each.

i. Links and Joints: Links are the solid structural members of a robot, and joints are the movable couplings between them.

ii. Degree of Freedom (dof): Degree of freedom is the number of independent movements a robot can realize with respect to its base. The number of axes is normally the same as the number of degrees of freedom of the robot. Each joint on the robot introduces a degree of freedom. Each degree of freedom can be a slider, rotary, or other type of actuator. Robots typically have five or six degrees of freedom as shown in Figure 13.7.

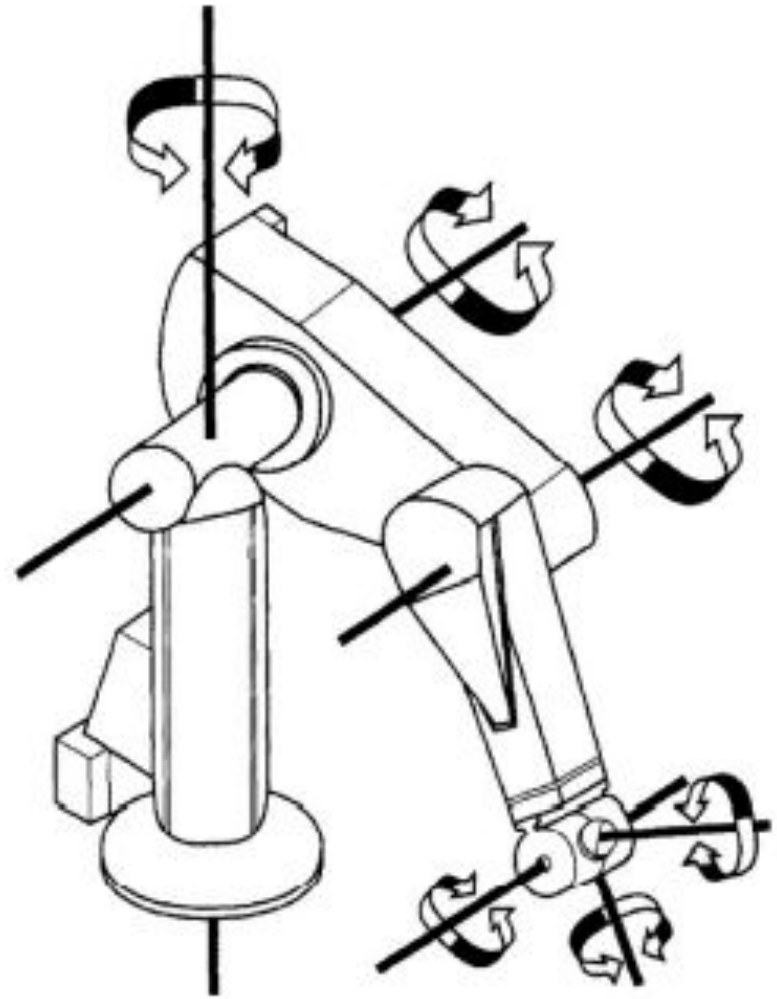


FIGURE 13.7 Robot with Six Degrees of Freedom.

- Figure 13.8 shows an industrial robot with three basic degrees of freedom plus three degrees of freedom in the wrist and a seventh in its ability to move back and forth along the floor.

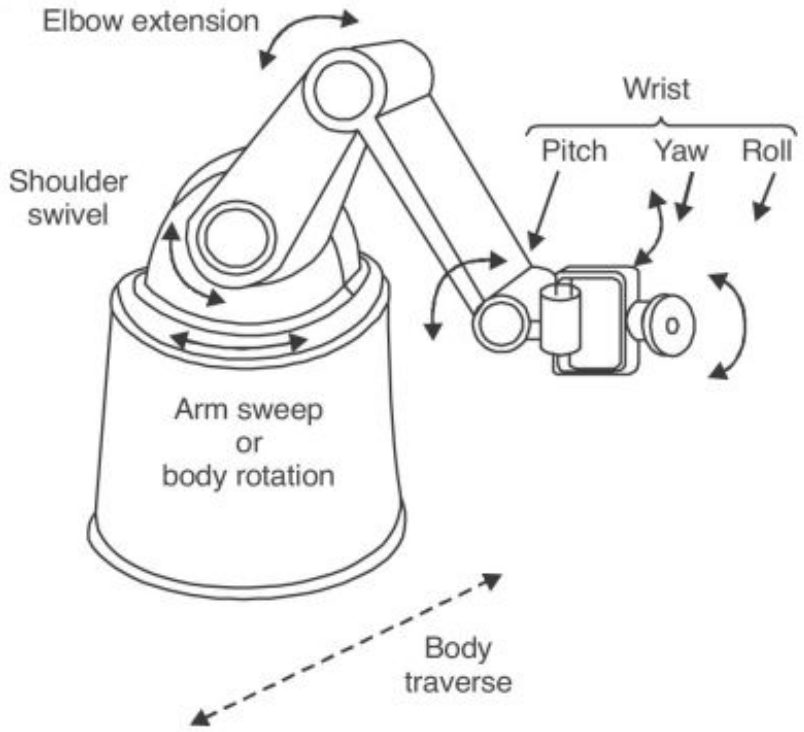
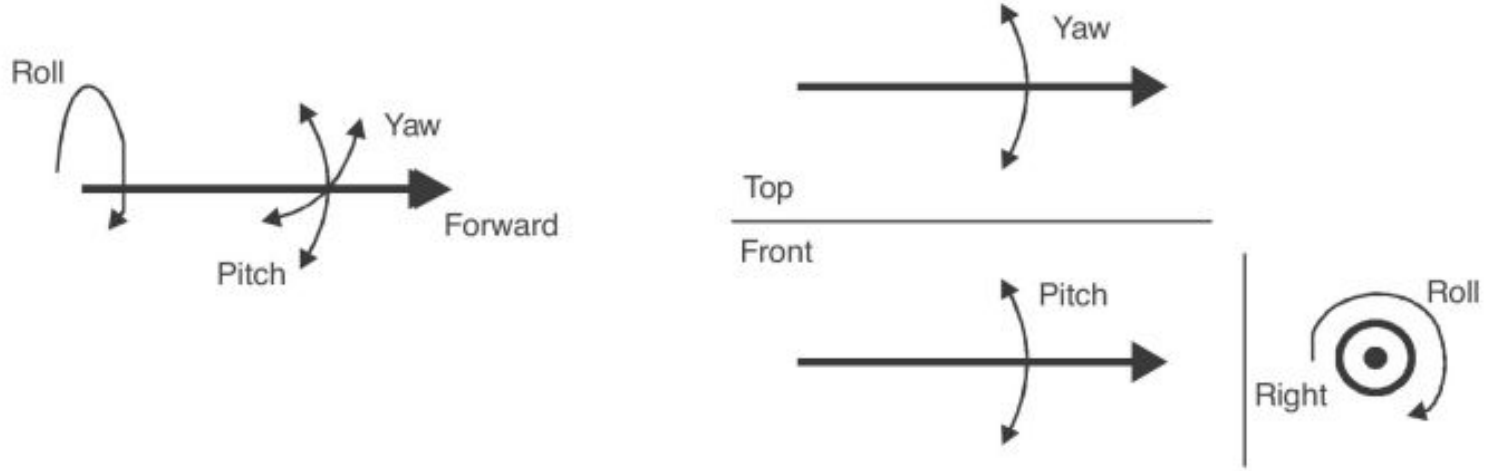


FIGURE 13.8 Robot with Seven Degrees of Freedom.

ROBOT TERMINOLOGY

iii)Orientation Axis: Basically, if the tool is held at a fixed position, the orientation determines which direction it can be pointed in. Roll, pitch, and yaw are the common orientation axes used.



ROBOT TERMINOLOGY

iv) Position Axis: The tool, regardless of orientation, can be moved to a number of positions in space.

v) Tool Center Point (TCP): The tool center point is located either on the robot, or the tool as shown in Figure 13.10. Typically, the TCP is used when referring to the robots position, as well as the focal point of the tool (e.g., the TCP could be at the tip of a welding torch). The TCP can be specified in Cartesian, cylindrical, spherical, etc., coordinates depending on the robot. As tools are changed, one often reprograms the robot for the TCP.

ROBOT TERMINOLOGY

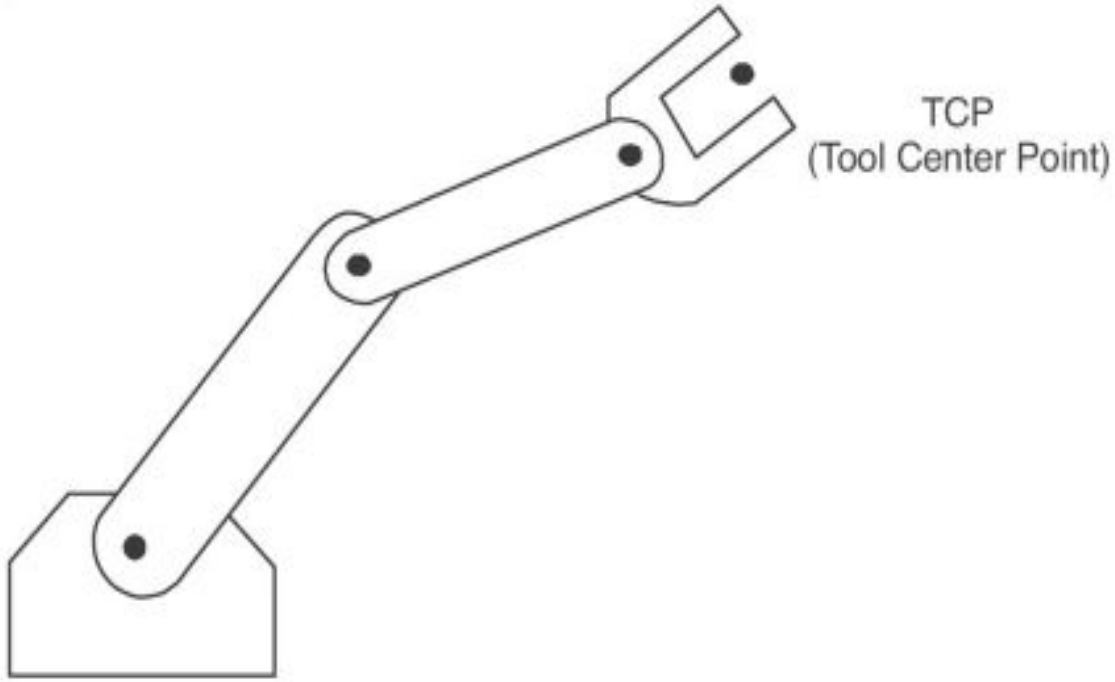


FIGURE 13.10 Tool Center Point.

ROBOT TERMINOLOGY

vi) Accuracy: Accuracy specification describes how close the arm will be when it moves to the desired point.

vii) Precision (validity): Precision is defined as how accurately a specified point can be reached. This is a function of the resolution of the actuators, as well as its feedback devices

viii)Repeatability (variability): Repeatability is how accurately the same position can be reached if the motion is repeated many times (Refer to Figure 13.11). Suppose that a robot is driven to the same point 100 times. Because many factors may affect the accuracy of the position, the robot may not reach the same point every time, but will be within a certain radius from the desired point.

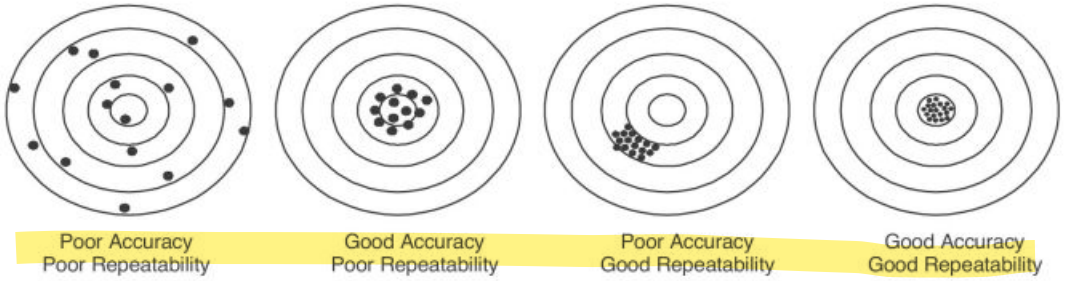


FIGURE 13.11 Accuracy versus Repeatability.

ROBOT TERMINOLOGY

xi) Work envelope/Workspace: A robot can only work in the area in which it can move. This area is called the work envelope. The work envelope is determined by how far the robot's arm can reach and how flexible the robot is. The more reach and flexibility a robot has, the larger the work envelope will be. It is one of the most important characteristics to be considered in selecting a suitable robot. Various robot configurations have different work envelopes. For a Cartesian configuration, the reach is a rectangular-type space.

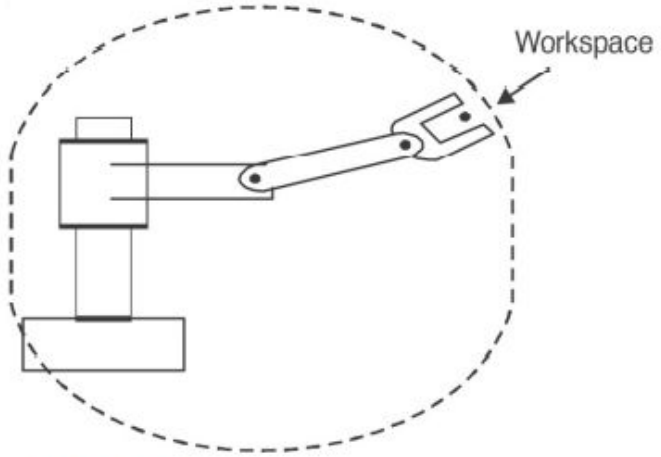


FIGURE 13.12 Work Envelope of a Robot.

ROBOT TERMINOLOGY

x) stability. Stability refers to robot motion with the least amount of oscillation. A good robot is one that is fast enough but at the same time has good stability.

xi) Speed. Speed refers either to the maximum velocity that is achievable by the Tool Center Point (TCP), or by individual joints. This number is not accurate in most robots, and will vary over the workspace as the geometry of the robot changes (and hence the dynamic effects). The number will often reflect the maximum safest speed possible. Some robots allow the maximum rated speed (100%) to be passed, but it should be done with great care.

ROBOT TERMINOLOGY

xii) Payload. Payload is the weight a robot can carry and still remain within its specifications. For example, a robot's maximum load capacity may be much larger than its specified payload, but at the maximum level, it may become less accurate, may not follow its intended path accurately, or may have excessive deflections. The payload of robots compared with their own weight is usually very small.

xiii)Reach. Reach is the maximum distance a robot can reach within its work envelope.

xiv) Settling Time. During a movement, the robot moves fast, but as the robot approaches the final position it slows down, and slowly approaches at final position. The settling time is the time required for the robot to be within a given distance from the final position.

ROBOTIC JOINTS

A robot joint is a mechanism that permits relative movement between parts of a robot arm. The joints of a robot are designed to enable the robot to move its end-effector along a path from one position to another as desired. The basic movements required for a desired motion of most industrial robots are:

- Rotational movement: This enables the robot to place its arm in any direction on a horizontal plane.
- Radial movement: This enables the robot to move its end-effector radially to reach distant points.
- Vertical movement: This enables the robot to take its end-effector to different heights.

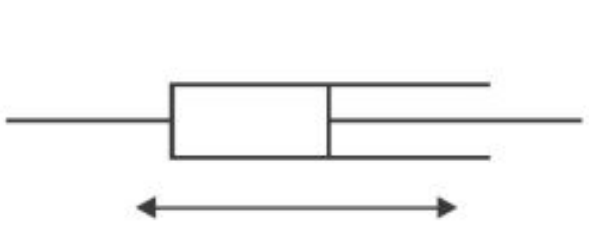
ROBOTIC JOINTS

Degrees of freedom, independently or in combination with others, define the complete motion of the end-effector. These motions are accomplished by movements of individual joints of the robot arm. The joint movements are basically the same as relative motion of adjoining links. Depending on the nature of this relative motion, the joints are classified as

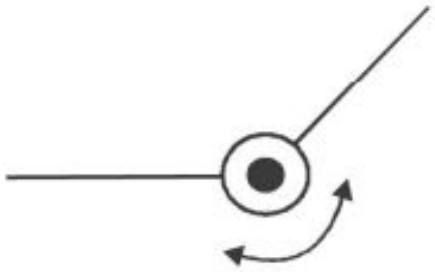
- Prismatic joints
- Revolute joints

ROBOTIC JOINTS

- Prismatic joints are also known as sliding as well as linear joints. They are called prismatic because the cross section of the joint is considered as a generalized prism. They permit links to make a linear displacement along a fixed axis. In other words, one link slides on the other along a straight line. These joints are used in gantry, cylindrical, or similar joint configurations.
- Revolute joints, the second type of joint is a revolute joint where a pair of links rotates about a fixed axis.



(a) Prismatic joint



(b) Revolute joint

ROBOTIC JOINTS

The variations of revolute joints shown in Figure 13.14 include: Rotational joint (R) Twisting joint (T) Revolving joint (V)

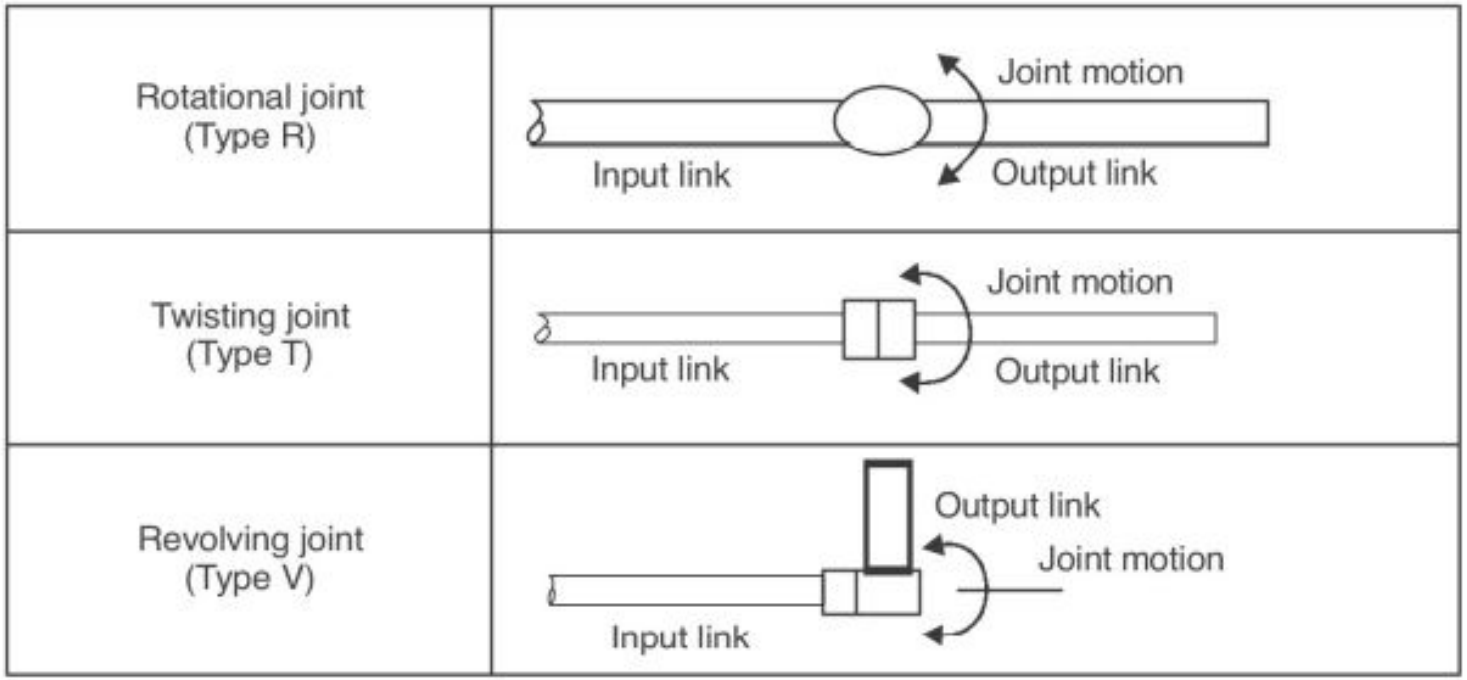


FIGURE 13.14 Revolute Joints.

ROBOTIC JOINTS

- Rotational joint (R) is identified by its motion, rotation about an axis perpendicular to the adjoining links. Here, the lengths of adjoining links do not change but the relative position of the links with respect to one another changes as the rotation takes place.
- A twisting joint (T) is also a rotational joint, where the rotation takes place about an axis that is parallel to both adjoining links.
- A revolving joint (V) is another rotational joint, where the rotation takes place about an axis that is parallel to one of the adjoining links. Usually, the links are aligned perpendicular to one another at this kind of joint. The rotation involves revolution of one link about another.

CLASSIFICATION OF ROBOTS

Robots can be classified by four fundamental elements of operation:
Coordinate systems

- Power source
- Method of control
- Programming method

ROBOT CLASSIFICATION ON THE BASIS OF COORDINATE SYSTEMS

Structurally, robots are classified according to the wrist's coordinate system as follows

- Cartesian/Rectilinear Robot: The axis or dimensions of these robots are 3 intersecting straight lines (x-y-z) as shown in Figure 13.15. The Cartesian coordinate robot is one that consists of a column and an arm. It is sometimes called an x-y-z robot, indicating the axis of motion. The x-axis is lateral motion, the y-axis is longitudinal motion, and the z-axis is vertical motion. Thus, the arm can move up and down on the z-axis; the arm can slide along its base on the xaxis; and then it can telescope to move to and from the work area on the y-axis. The features of a Cartesian robot (electronic equipment, control program) are same as those of CNC machine tools

Cartesian/Rectilinear Robot

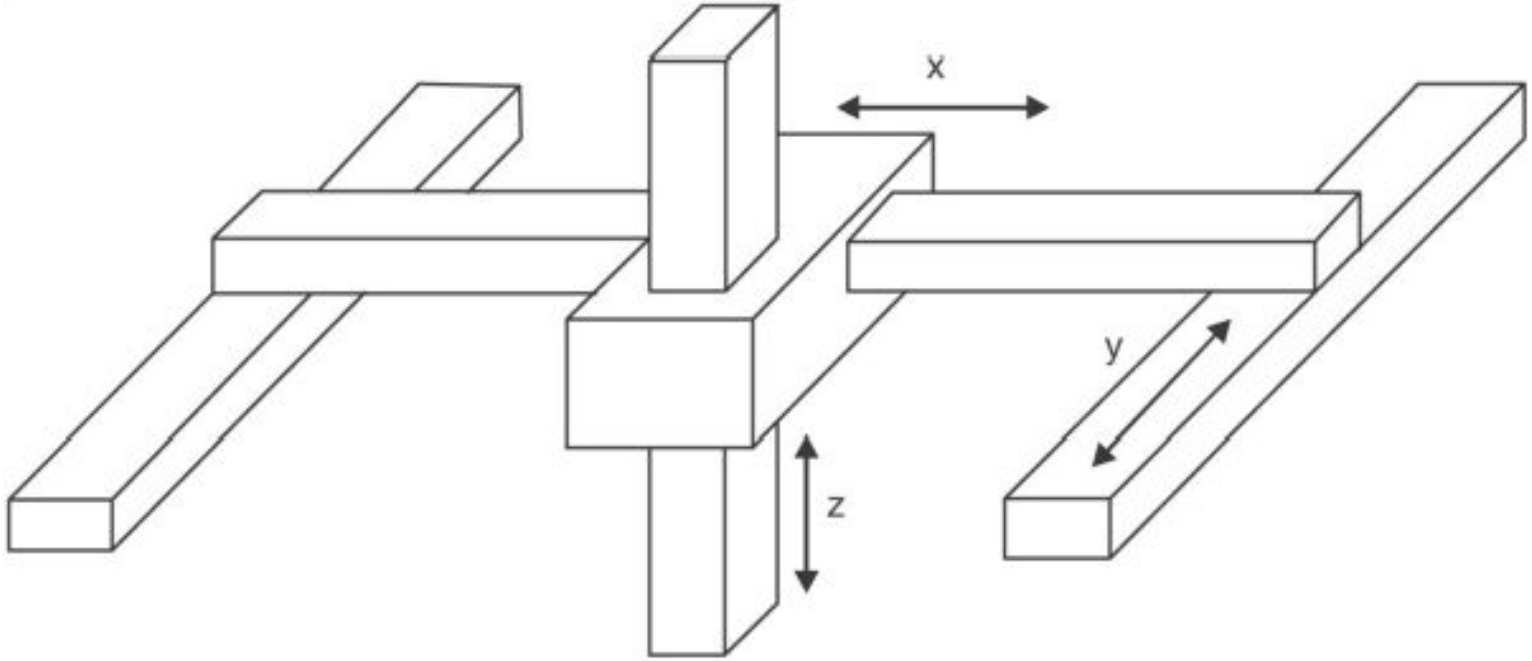


FIGURE 13.15 Cartesian Robot.

Cartesian/Rectilinear Robot

Applications

- These types of robots are used for:
- Pick and place work
- Application of sealant
- Assembly operations Handling machine tools
- Arc welding

Advantages

- Ability to do straight line insertions into furnaces
- Easy computation and programming
- Most rigid structure for given length

Disadvantages

- Requires large operating volume
- Exposed guiding surfaces require covering in corrosive or dusty environments Can only reach front of itself
- Axis hard to seal

Cylindrical Robot

- Cylindrical robots have one angular dimension and 2 linear dimensions as shown in Figure 13.16. The rigid structure of this system offers them the capability to lift heavy loads through a large working envelope. The main body of such a robot consists of a horizontal arm mounted on a vertical column. The column is mounted on a rotating base. The horizontal arm moves forward and backward on the direction of the longitudinal axis and it also moves up and down on the column. Column and arm are rotating on the base around the vertical axis.

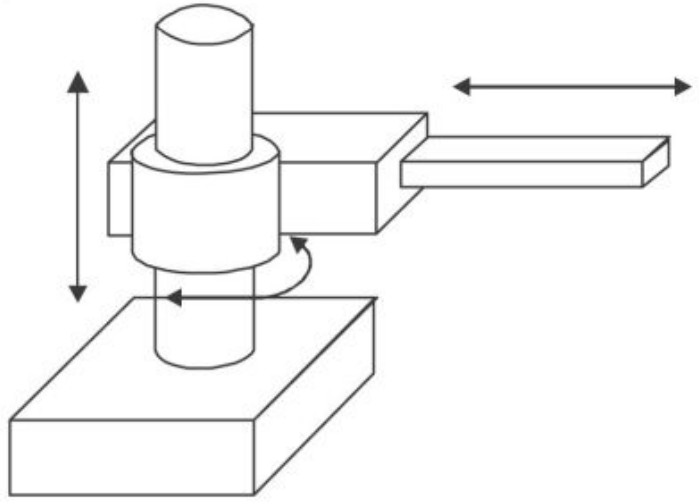


FIGURE 13.16 Cylindrical Robot.

Cylindrical Robot

Applications

- Assembly operations
- Handling machine tools
- Spot-welding
- Handling die-casting machines

Advantages

- Can reach all around itself
- Rotational axis easy to seal
- Relatively easy programming
- Rigid enough to handle heavy loads through large working space
- Good access into cavities and machine openings

Disadvantages

- Can’t reach above itself
- Won’t reach around obstacles
- Exposed drives are difficult to cover from dust and liquids Linear axes is hard to seal

Spherical (Polar) Robot

Robots of this type consist of a rotating base, a lifting part, and a telescopic arm, which moves inwards and outwards (refer to Figure 13.17). The 2 dimensions of spherical robots are angles and the third is a linear distance from the point of origin. These robots operate according to spherical coordinates and offer greater flexibility. This design is used where a small number of vertical actions are adequate

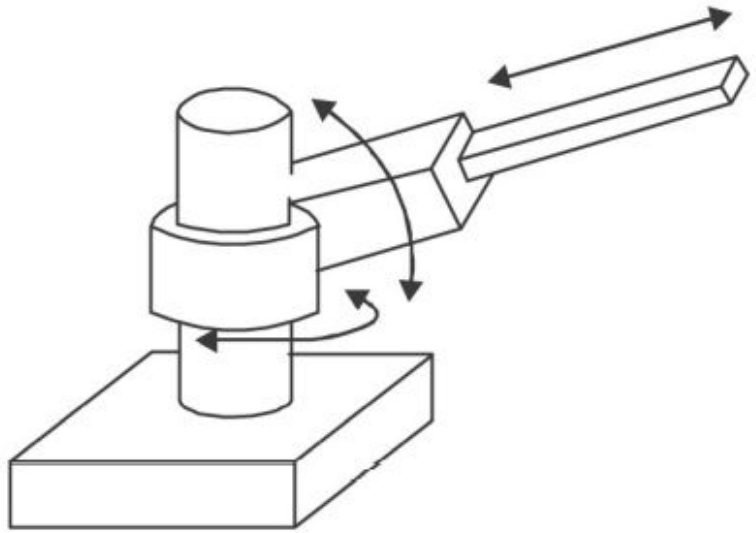


FIGURE 13.17 Spherical Robot.

Spherical (Polar) Robot

Applications

- Handlings at die casting or fettling machines
- Handling machine tools
- Arc/spot welding

Advantages

- Large working envelope
- Two rotary drives are easily sealed against liquids/dust

Disadvantages

- Complex coordinates more difficult to visualize, control, and program
- Exposed linear drive
- Low accuracy

Articulated Robot

Articulated robots consist of three constant parts (links) that are joined with revolute joints and are placed on a rotating base as shown in Figure 13.18. The kinematics layout is similar to a human arm. The tool (gripper) is similar to a palm and is adjusted to the lower part of the arm through the wrist. The elbow connects the lower and upper part of the arm and the shoulder connects the upper part of the arm with the base. Many times the shoulder joint has a rotational motion in the horizontal plane. The articulated robot has all three axes revolute, so the position resolution is completely dependent on the arm's position. The total accuracy of an articulated robot is small, because joint errors are accumulated at the end of the arm, which is at the wrist position.

Articulated Robot

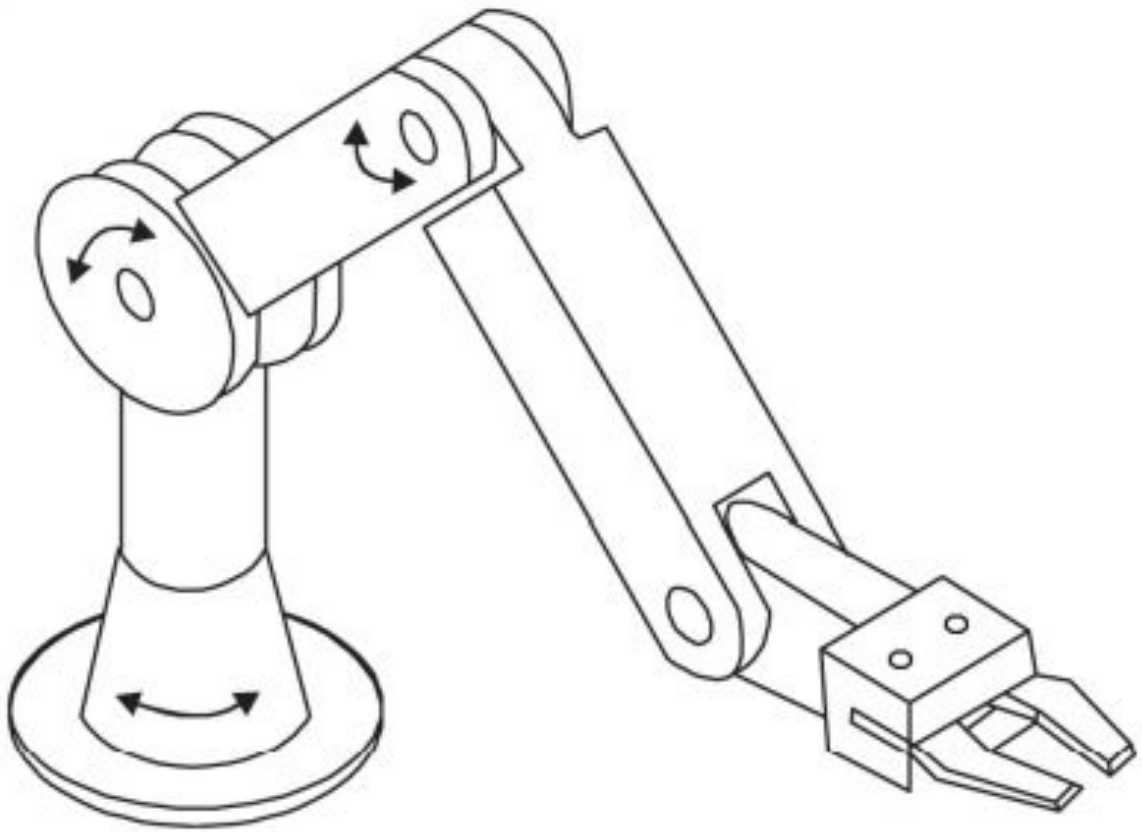


FIGURE 13.18 Articulated Robot.

Articulated Robot

Applications

- Assembly operations
- Die casting
- Fettling machines
- Gas welding
- Arc welding
- Spray-painting

Advantages

- High mechanical flexibility
- They can move with high speed at three degrees of freedom
- All joints can be sealed from the environment

Disadvantages

- Extremely difficult to visualize, control, and program
- Restricted volume coverage
- Low accuracy

SCARA Robot

One style of robot that has recently become quite popular is a combination of the articulated arm and the cylindrical robot. This robot has more than three axes and is called a SCARA robot (refer to Figure 13.19). It is used widely in electronic assembly. The rotary axes are mounted vertically rather than horizontally. This configuration minimizes the robot's deflection when it carries an object while moving at a programmed speed. The acronym SCARA stands for Selective Compliance Assembly Robot Arm, a particular design developed in the late 1970s.

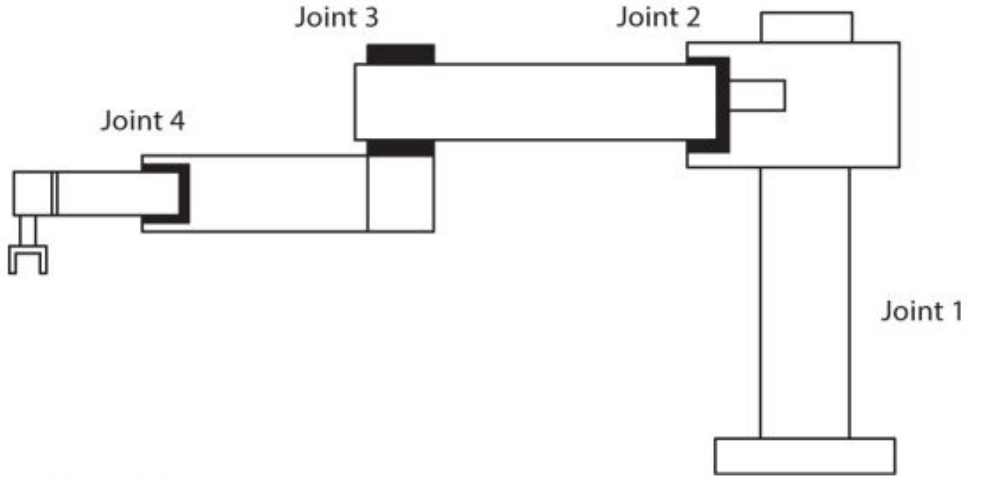


FIGURE 13.19 SCARA Robot.

SCARA Robot

Applications

- Pick and place work
- Assembly operations
- Application of sealant
- Handling machine tools

Advantages

- High speed Excellent repeatability
- Good payload capacity
- Large work area for floor space
- Moderately easy to program

Disadvantages

- Limited application
- Two ways to reach point
- Difficult to program off-line
- Highly complex arm

ROBOT CLASSIFICATION ON THE BASIS OF POWER SOURCE

Actuators drive the mechanical linkages and joints of a manipulator, which can be various types of motors and valves. The energy for these actuators is provided by some power source such as hydraulic, pneumatic, or electric. There are three major types of drive systems for industrial robots:

- Hydraulic drive system
- Pneumatic drive system
- Electric drive system

Hydraulic Drive System

- The most popular form of the drive system is the hydraulic drive system because hydraulic cylinders and motors are compact and allow for high levels of force and power, together with accurate control. These systems are driven by a fluid that is pumped through motors, cylinders, or other hydraulic actuator mechanisms. A hydraulic actuator converts forces from high pressure hydraulic fluid into the mechanical shaft rotation or linear motion. Hydraulic robots are preferred in environments in which the use of electric drive robots may cause fire hazards, for example, in spray painting.

Advantages

- A hydraulic device can produce an enormous range of forces without the need for gears, simply by controlling the flow of fluid Preferred for moving heavy parts
- Preferred to be used in explosive environments Self-lubrication and selfcooling
- Smooth operation at low speeds
- There is need for return line

Disadvantages

- Occupy large space area
- There is a danger of oil leak to the shop floor

Pneumatic Drive System

- Pneumatic drive systems are found in approximately 30 percent of today's robots. These systems use compressed air to power the robots. Because machine shops typically have compressed air lines in their working areas, the pneumatically driven robots are very popular. These robots generally have fewer axis of movement and can carry out simple pick-andplace material-handling operations, such as picking up an object at one location and placing it at another location. These operations are generally simple and have short cycle times. The pneumatic power can be used for sliding or rotational joints.

Advantages

- Less expensive than electric or hydraulic robots Suitable for relatively less degrees of freedom design
- Do not pollute work area with oils
- No return line required Pneumatic devices are faster to respond as compared to a hydraulic system as air is lighter than fluid

Disadvantages

- Compressibility of air limits control and accuracy aspects
- Noise pollution from exhausts Leakage of air can be of concern

Electric Drive System

Electrical drive systems are used in about 20 percent of today's robots. These systems are servomotors, stepping motors, and pulse motors. These motors convert electrical energy into mechanical energy to power the robot. Compared with a hydraulic system, an electric system provides a robot with less speed and strength. Electric drive systems are adopted for smaller robots. Electrically driven robots are the most commonly available and used industrial robots.

There are three major types of electric drive that have been used for robots:

- Stepper Motors: These are used mainly for simple pick and place mechanisms where cost is more important than power or controllability.
- DC Servos: For the early electric robots, the DC servo drive was used extensively. It gave good power output with a high degree of control of (c) both speed and position.
- AC Servos: In recent years, the AC servo has taken over from the DC servo as the standard drive. These modern motors give higher power output and are almost silent in operation. As they have no brushes, they are very reliable and require almost no maintenance in operation.

Advantages

- Good for small and medium size robots
- Better positioning accuracy and repeatability Less maintenance and reliability problems

Disadvantages

- Provides less speed and strength than hydraulic robots
- Not all electric motors are suited for use as actuators in robots
- Require more sophisticated electronic controls and can fail in high temperature, wet, or dusty environments

ROBOT CLASSIFICATION ON THE BASIS OF METHOD OF CONTROL

- The motions of a robot are controlled by a combination of software and hardware that is programmed by the user. Robots are classified by control method into servo and non-servo robots.

Non-Servo Controlled Robots

- Non-servo control is a purely mechanical system of stops and limit switches, which are pre-programmed for specific repetitive movements.
- This can provide accurate control for simple motions at low cost. The motions of non-servo controlled robots are controlled only at their endpoints, not throughout their paths. Non-servo robots are often referred as “endpoint,” “pick and place,” or “limited sequence” robots. These robots are used primarily for materials transfer.

Characteristics of non-servo robots include:

- Relatively high speed possible due to smaller size of the manipulator
- These robots are low cost and simple to operate, program, and maintain
- These robots have limited flexibility in terms of program capacity and positioning capability

Servo Controlled Robots

Servo control system is capable of controlling the velocity, acceleration, and path of motion, from the beginning to the end of the path. It uses complex control programs. These systems use programmable logic controllers (PLCs) and sensors to control the motions of robots. They are more flexible than non-servo systems, and they can control complicated motions smoothly. Sensors are used in servo-control systems to track the position of each of the axis of motion of the manipulator. Servo controlled robots are classified according to the method that the controller uses to guide the end-effector. These are:

- Point-to-point (PTP)
- control robot Continuous-path (CP)
- control robot Controlled-path robot

Point-to-Point Control Robot (PTP)

- A point-to-point robot is capable of moving from one discrete point to another within its working envelope.
- During point-to-point operation the robot moves to a position, which is numerically defined, and it stops there.
- The end effector performs the desired task, while the robot is halted. When task is completed, the robot moves to the next point and the cycle is repeated.

Applications:

- Component insertion
- Spot-welding
- Hole drilling
- Machine loading and unloading
- Assembly operations

Continuous-path (CP) Control Robot

- In a continuous-path robot, the tool performs its task, while the robot (its axes) is in motion, like in the case of arc welding, where the welding pistol is driven along the programmed path. All axes of continuous path (c) robots move simultaneously, each with a different speed. The computer coordinates the speeds so that the required path is followed. The robot's path is controlled by storing a large number of spatial points in the robot's memory during the teach sequence

Applications:

- Spray painting
- Finishing
- Gluing Arc welding operations
- Cleaning of metal articles
- Complex assembly processes

Controlled-path Robot

- In controlled-path robots, the robot is moved along a computer generated, predictable path as the robot travels from point to point.
- The computer-generated path may be a straight line with end-effector orientation or it may involve curved paths through successive points and/or gradual orientation changes.
- Good accuracy can be obtained at any point along the specified path. Only the start and finish points and the path definition function must be stored in the robot's control memory.
- The robot movements are more precise than with point-to-point programming and are less likely to present a hazard to personnel and equipment.

ROBOT CLASSIFICATION ON THE BASIS OF PROGRAMMING METHOD

- Manual programming
- Lead-through programming
- Walk-through programming

ROBOT SELECTION

Once the application is selected, a suitable robot should be chosen from the many commercial robots available in the market. The characteristics of robots generally considered in a selection process include:

- Size of class
- Degrees of freedom
- Velocity Drive type Control mode
- Repeatability
- Lift capacity
- Right-left,
- Up-down, and In-out traverse Yaw, pitch, and roll
- Weight of the robot

ROBOT SELECTION

1. Size of class: The size of the robot is given by the maximum dimension (x) of the robot work envelope.

- Micro ($x < 1$ m) Small (1 m $< x < 2$ m)
- Medium (2 m $< x < 5$ m)
- Large ($x > 5$ m)

2. Degrees of freedom: The cost of the robot increases with the number of degrees of freedom. Six degrees of freedom is suitable for most works.

3. Velocity: Velocity consideration is effected by the robot's arm structure.

- Rectangular
- Cylindrical
- Spherical
- Articulated

ROBOT SELECTION

4. Drive type:
- Hydraulic
- Electric
- Pneumatic

5. Control mode:
- Point-to-point control (PTP)
- Continuous path control (CP)
- Controlled path control

6. Lift capacity:
- 0–5 kg
- 5–20 kg
- 20–40 kg and so forth